

Alwin Philip

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EXPERIENCE

Software Development Specialist (Analytics/AI) <i>New Jersey Department of Labor and Workforce Development, Trenton, NJ</i>	July 2026 – Present
● Build interactive analytics dashboards using AWS QuickSight connected to the department's AWS Data Lake, surfacing workforce and labor market trends for leadership and program staff. ● Develop and maintain AI/ML applications using AWS Bedrock, including prompt engineering workflows for generative AI integration across department operations.	
Agentic AI Engineer (Contract) <i>Beacon Health AI</i>	May 2026 – Present
● Build and analyze automated agentic workflows for document processing, including an automated fax-filing agent, using Cyberdesk as the authoritative data source for run-history tracking. ● Collaborate with a cross-functional engineering team to evaluate data infrastructure reliability and resolve discrepancies in agent run-history logging.	
Graduate Research Assistant – Systems and Data Engineering <i>Rutgers University–Camden, Camden, NJ</i>	December 2025 – May 2026
● Engineered a production-grade automated ingestion pipeline across a 27-year SEC EDGAR archive — Python and Shell scripts handling ingestion, schema transformation, and structured output for 10-K, 8-K, and 6-K filings. ● Wrote Amarel HPC batch automation scripts eliminating manual preprocessing across large-scale financial and behavioral datasets.	
Marketing Analytics and AI Specialist <i>Rutgers–Camden School of Business, Camden, NJ</i>	August 2024 – May 2026
● Led GitHub Copilot and Microsoft Copilot rollout at Thayer Distribution — measured 20 percent delivery velocity improvement post-deployment. ● Deployed and maintained Google Looker Studio and GA4 analytics dashboards as production tools in active daily use by organizational leadership.	
Data Science and AI Research Intern <i>Code-blue.ai (UC Berkeley-Affiliated Startup), Berkeley, CA</i>	June 2025 – August 2025
● Designed and delivered real-time stroke detection systems end-to-end — feature extraction, high-volume facial/speech stream ingestion, and inference pipeline — achieving 98.9 percent model accuracy at production scale.	

PROJECTS

Core Sentinel – Windows AI Security Agent (Capstone)	January 2026 – Present
● Architected a Windows background service (pywin32) with a locally deployed ONNX-quantized TinyBERT model — under 100ms latency, under 5% CPU, ~50MB RAM, zero data exfiltration. Built a PyQt6 overlay UI with a no-code policy engine covering FERPA, HIPAA, and SEC domains.	
Arch Street Burial Site Project	March 2026 – May 2026
● Built an interactive AI/ML dashboard analyzing 324 unnamed burials from a historic Philadelphia cemetery — 3D excavation visualization (Three.js), DBSCAN clustering, and Bayesian identity-confidence matching. Deployed via React with CI/CD on Netlify as part of Rutgers–Camden's inaugural AI Campus Cohort.	
Multimodal Dementia Classification System	December 2025
● Trained and shipped an end-to-end ML system: audio ETL via OpenAI Whisper, multimodal fusion model achieving 90.2% classification accuracy, Captum XAI explainability, and ONNX export for framework-agnostic deployment.	
CVLens – Full-Stack AI Recruitment Application	September 2025
● Architected a full-stack AI application from zero to production: Python NLP backend, Firestore NoSQL data layer, n8n workflow orchestration, and candidate ranking API.	

EDUCATION

Rutgers University <i>M.S. Computer Science GPA: 3.65 President, Graduate Student Organization</i>	Camden, NJ September 2024 – May 2026
Amity University <i>B.Tech Information Technology GPA: 3.82</i>	Noida, India September 2020 – June 2024

TECHNICAL SKILLS

Languages: Python (primary), C++, JavaScript, SQL, Shell Scripting
AI/ML: NLP, LLMs, ONNX Runtime, OpenAI Whisper, Captum XAI, TensorFlow, PyTorch, Scikit-learn, Agentic AI Systems
Systems & Backend: pywin32, PyQt6, REST APIs, ETL Pipelines, n8n, Docker, Git, Amarel HPC
Cloud & Data: AWS (QuickSight, Bedrock), AWS SAA (In Progress), GCP, Firestore, MongoDB, SQL, SEC EDGAR

RESEARCH, PUBLICATIONS & CERTIFICATIONS

Behavioral Biometrics and Phishing Detection — Ongoing — hybrid fraud detection model (XGBoost, Random Forest, Neural Networks), 45% improvement over behavioral-only baselines	
Quantum Networking Security — IEEE ICRITO 2024 — Best Paper Award	
Digital Avatars and Metaverse Security — International Journal of Systems and Applications, Nov 2024	
● AWS Certified Solutions Architect – Associate (Expected September 2026) Google Analytics Advanced Google Cloud Professional: Generative AI, Security, and Infrastructure Modernization (2024) ● Cisco CCNA and CyberOps Associate Core Leadership Certification – Rutgers School of Business (2025) 1st Place – TECHNOVATE 2024	